Problem 1.14

Suppose you deposit \$1000 into a savings account that pays annual interest rate of 0.4% compounded quarterly.

1. What is the balance in the account at the end of year?
2. What is the interest earned over the year period?
3. What is the effective interest rate?

Solution

For a nominal annual interest rate r compounded n times per year, the accumulated amount after t years is

$$A = P\left(1 + \frac{r}{n}\right)^{nt}.$$

Here,

$$P = 1000, \qquad r = 0.004, \qquad n = 4, \qquad t = 1.$$

(a) Balance at the end of the year

$$A = 1000\left(1 + \frac{0.004}{4}\right)^{4}.$$

Thus,

$$A = 1000(1.001)^4 \approx 1004.006.$$

So the balance is

$$\boxed{\$1004.006 \approx \$1004.01}.$$

(b) Interest earned over the year

$$\text{Interest} = 1004.006 - 1000 = 4.006.$$

Therefore,

$$\boxed{\$4.006 \approx \$4.01}.$$

(c) Effective interest rate

The effective annual interest rate is

$$\left(1 + \frac{r}{n}\right)^{n} - 1.$$

So,

$$\left(1 + \frac{0.004}{4}\right)^{4} - 1 = (1.001)^4 - 1 \approx 0.004006.$$

Hence, the effective interest rate is

$$\boxed{0.4006\% \approx 0.401\%}.$$